

Muhammad Tayyab

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[LinkedIn](#) | [GitHub](#)

SUMMARY

AI Engineer specializing in the development and deployment of end-to-end Machine Learning, Deep Learning, and Computer Vision solutions. Proven expertise in architecting **RAG systems**, performing **LLM fine-tuning**, and optimizing models for real-time inference. Award-winning developer with a track record of building scalable AI products, including an AI-powered fitness startup and high-accuracy exoplanet classification systems. Proficient in **Python**, **TensorFlow**, and **PyTorch**, with a strong academic background (3.6 CGPA) and experience bridging the gap between complex ML models and production-ready applications

PROJECTS

Fitro-AI-Powered Fitness App (Startup - NIC Hyderabad Cohort 7)

[FITRO](#)

Problem: In Pakistan, millions struggle with calorie tracking due to limited health literacy and app fatigue.

Solution: Built a mobile app using computer vision to detect food calories from images and use LLMs for real-time fitness guidance.

Impact: Enabled >90% faster logging of meals vs. manual entry. API-first LLM setup ensures affordability and scalability across devices

ExoQuark | AI-Powered Exoplanet Classification System

[Live Demo](#)

[GitHub](#)

Technologies: Next.js, TypeScript, Three.js, AWS Bedrock, Flask, XGBoost

Problem: Astronomical datasets from Kepler, K2, and TESS showed >30% false positives in exoplanet classification.

Solution: Engineered ExoQuark, an AI ensemble (XGBoost + LightGBM) achieving ~89% classification accuracy across satellites with calibrated thresholds and explainable outputs.

Impact: Cut false detections by 40%, boosting reliability of exoplanet research through an open-access platform for scientists.

Health Mate | AI-Powered Medical Analyzer

[Live Demo](#)

[GitHub](#)

Technologies: Next.js 14, TypeScript, Google Gemini 2.5 Flash, Supabase, Clerk

Developed an AI-driven diagnostic tool during the SMIT Coding Night Hackathon to analyze medical reports (PDFs, X-rays).

Integrated Google Gemini 2.5 Flash to generate instant bilingual summaries (English + Roman Urdu) for patients.

Education

Mehran University of Engineering and Technology

Bachelor of Science in Artificial Intelligence (BSAI); **CGPA: 3.60**

Computer Man Institute of Emerging Technologies

Diploma in Information Technology (DIT);

Jamshoro, Sindh

July 2023 - Aug 2027

Hyderabad, Sindh

Oct 2021 - August 2022

Awards & Honors

Winner, Technofest 2026 – ML Mastery Competition

Built and optimized a Machine Learning model to predict life expectancy, achieving a R-squared score of 0.97. Demonstrated strong skills in data preprocessing, feature engineering, model evaluation, and performance optimization under competitive conditions.

Winner, NASA Space Apps Challenge Hyderabad (2025)

Recognized for building "ExoQuark," an AI/ML space exploration tool. Nominated for Global Round.

Winner, AWS MUET Hacktoberfest (2024)

Developed a cross-platform mobile application using Expo React Native.

WORK EXPERIENCE

Web Development Intern | MUET FM 92.6

[MUET FM 92.6](#)

Spearheaded full-stack development of the official website, migrating legacy systems to a modern React architecture.

Optimized frontend performance, reducing page load times by **40%** and boosting SEO rankings through server-side rendering (SSR) best practices.